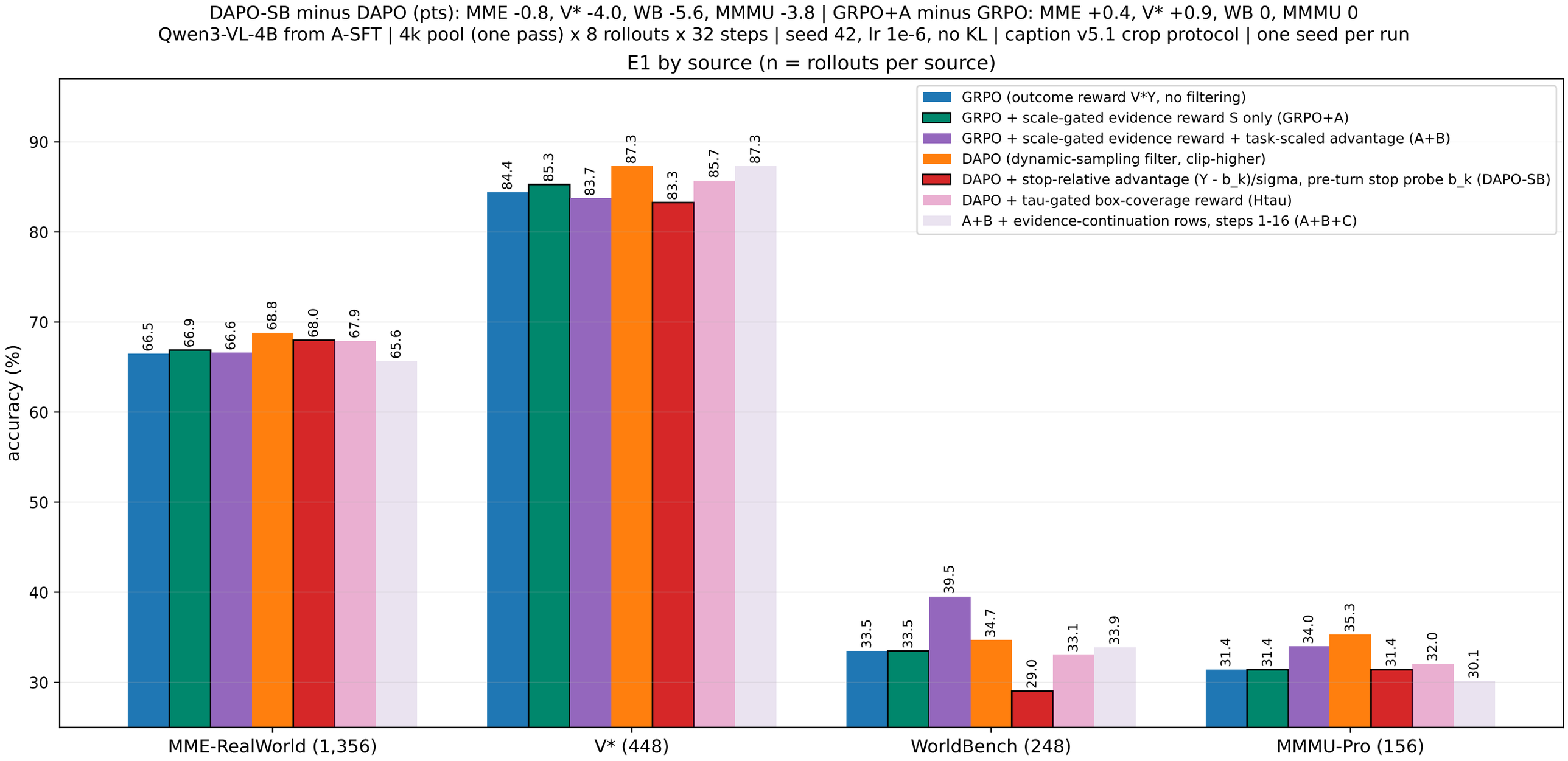
DAPO-SB minus DAPO (pts): MME -0.8, V* -4.0, WB -5.6, MMMU -3.8 | GRPO+A minus GRPO: MME +0.4, V* +0.9, WB 0, MMMU 0
Qwen3-VL-4B from A-SFT | 4k pool (one pass) x 8 rollouts x 32 steps | seed 42, lr 1e-6, no KL | caption v5.1 crop protocol | one seed per run
E1 by source (n = rollouts per source)
GRPO (outcome reward V*Y, no filtering)
GRPO + scale-gated evidence reward S only (GRPO+A)
GRPO + scale-gated evidence reward + task-scaled advantage (A+B)
DAPO (dynamic-sampling filter, clip-higher)
DAPO + stop-relative advantage (Y - b_k)/sigma, pre-turn stop probe b_k (DAPO-SB)
DAPO + tau-gated box-coverage reward (Htau)
A+B + evidence-continuation rows, steps 1-16 (A+B+C)
accuracy (%)
90
80
70
60
50
40
30
66.5
66.9
66.6
68.8
68.0
67.9
65.6
84.4
85.3
83.7
87.3
83.3
85.7
87.3
33.5
33.5
39.5
34.7
29.0
33.1
33.9
31.4
31.4
34.0
35.3
31.4
32.0
30.1
MME-RealWorld (1,356)
V* (448)
WorldBench (248)
MMMU-Pro (156)